KETCo Unified Digital Platform

Requirements & Business Architecture Specification Document

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1. Project Overview & Scope

1.1 Profile of the Organization

The Kerry Environmental Testing Cooperative (KETCo) operates a central laboratory in Tralee, County Kerry. KETCo tests water and soil samples for rural and local stakeholders. Clients give samples directly to the Tralee laboratory. Alternatively, KETCo technicians collect samples in the field.

Clients request tests for three reasons:

- General environmental monitoring.
- Specific project work.
- Investigations after unexpected environmental changes.

1.2 Current Process

Intake and Logging: Staff use paper forms, spreadsheets, and emails. They manually record requests, labels, field data, and test assignments.

Collection: Field technicians, clients, or partner organizations collect samples.

Receipt and Screening: Staff check samples at the intake desk. They look for leaks, label errors, temperature changes, and low volume. Staff write a laboratory reference number on the container.

Internal Routing: Staff move samples between laboratory stations. They track steps with paper bench sheets.

Testing and Referrals: The manager assigns tests. The manager checks machine availability and analyst training. Analysts split samples or send fractions to external laboratories.

Review and Reporting: Analysts record raw measurements. They calculate data manually. A reviewer must verify the data before report release. Staff use a manual workflow for report corrections.

1.3 Core Business Problems

High Administration Workload: Staff write the same data many times across paper logs and spreadsheets. This creates workflow bottlenecks.

Data and Entry Errors: Staff use different names for the same location. Handwritten labels are frequently illegible.

Custody Log Failures: Paper bench sheets make audit tracking difficult. Staff mix up sample bottles within the same batch.

Sample Loss Risk: The laboratory does not have a digital check for container types, preservation states, and holding times. Staff analyze degraded samples, which creates invalid data.

Security and Report Vulnerabilities: Customers can see unverified provisional values before review. A lack of data isolation risks exposing sensitive data to competitors.

1.4 Expected Outcomes

Unified Digital Platform: Replace paper and spreadsheets with one central software application.

Digital Chain of Custody: Create a permanent digital log from collection to disposal to stop bottle mix-ups.

Standard Locations: Separate broad text descriptions from precise GPS coordinates in the database.

Real-Time Visibility: Give the manager a live dashboard for intake checks, active tests, machine delays, and pending reports.

Versioned Reports: Transmit secure reports. Keep a full history of version numbers, revision notes, and recipient logs.

1.5 Known Constraints

- The system manages only water and soil samples.
- The internet is stable in the Tralee office but weak at rural collection points.
- One request can have many samples, and one sample can have many tests.
- Split fractions must link to the parent sample reference number.
- Machine calibration and Quality Control (QC) check failures delay testing.
- Unusual results are data anomalies. They do not prove environmental pollution or causes.

2. Stakeholder and Actor Catalogue

2.1 Stakeholder Profiles

Direct Stakeholders (Laboratory Staff): Receptionists, analysts, and managers must reduce double-entry workload. They must avoid bottle mix-ups.

Indirect Stakeholders (KETCo Executive Board): The board must decrease liability risks from unverified provisional data releases. They must stop competitors from viewing multi-tenant data.

External Stakeholders (Clients): Rural farmers and commercial entities require reliable and fast turnarounds. They demand total isolation of their sensitive commercial environmental profiles.

Partner Organizations: Partners require clear and simple drop-off field procedures under weak network conditions.

2.2 Actor-Goal Catalogue

Actor Name	Type	Description	Core System Goals
Reception Technician	Primary	Staff managing intake desks.	Generate unique IDs. Run baseline rule checks. Assign provisional status safely.
Field Technician	Primary	Remote operator collecting physical samples.	Log collection metrics offline. Capture stable coordinates linked to parent location profiles.
Analyst	Primary	Lab specialist executing technical assays.	Log numeric measurements. Flag departures. Execute instrument validations safely.
Quality Manager	Primary	Certified data reviewer and auditor.	Verify raw data metrics. Sign off version 1 records. Process version-controlled replacements.
Laboratory Manager	Primary	System supervisor monitoring operations.	Monitor bottlenecks via a dynamic console. Oversee active delays and anomalies.
Customer Portal	Primary	External representative interacting via web UI.	Query secure chronological version stacks. Download validated reports safely.
Notification Subsystem	Supporting	Internal engine distributing data alerts.	Distribute SMS/Email flags automatically within 5 minutes of critical failures.

3. System Boundaries

3.1 In-Scope Functions

Intake: Record customer requests, track sample quality metrics, and manage provisional acceptance.

Mapping: Separate broad locations from precise GPS coordinates in the database.

Custody Log: Track sample movement, handler IDs, station changes, and bench sheet references with timestamps.

Architecture: Link parent samples to internal splits and external laboratory records.

Management: Assign tests by analyst training certificates, track test status, and record raw measurements.

Instruments: Log machine availability, calibration data, and pre-test QC checks.

Reporting: Create final reports with sequential version numbers, mandatory notes, and access control.

3.2 Out-of-Scope Functions

Machine Connections: No direct hardware APIs or drivers. Analysts must enter all machine data manually.

Regulatory Compliance: No evaluation of statutory limits. No automated alerts to government authorities.

Accounting: No processing of client invoices, billing data, or financial workflows.

Consultation: No automated environmental conclusions or professional advice.

4. Dependencies and Information Gaps

4.1 External Dependencies

Rural Networks: Mobile operations depend on local network coverage or offline data caching.

Partner Accuracy: Early tracking depends on partner organizations recording field data correctly.

Third-Party Labs: Final reports depend on the delivery schedules of external laboratories.

4.2 Critical Information Gaps

Provisional Rules: Criteria for provisional acceptance and maximum holding times before system rejection.

Rejection Matrix: Rules that map physical defects to automatic test unsuitability.

Tracking Standards: Naming standards for internal splits and external referral samples.

Privilege Matrix: Definitions of roles that can approve method departures, override failed QC checks, or start re-tests.

SLA Logic: System logic to recalculate completion dates when machines fail.

Correction Governance: Review workflows and clearances to approve and distribute replacement reports.

Data Isolation: Access rules to restrict view permissions by customer type.

5. Requirements and Business Rules

5.1 Requirements Catalogue

ID	Subsystem	Name	Description	Pri	Rationale / Verification
FR-101	Intake	Unique ID Gen	The system must generate a unique reference ID for each sample at receipt.	Must	**Rationale:** Fundamental operational dependency. **Verify:** Inspection of data rows.
FR-102	Intake	Spatial Mapping	The system must record GPS coordinates and link them to a parent location ID.	Must	**Rationale:** Solves geographical naming ambiguity. **Verify:** Entry mapping demo.
FR-103	Intake	Cond. Status	Let technician set status to 'Provisionally Accepted' and force note entry.	Must	**Rationale:** Handle incomplete field data safely. **Verify:** Null note rejection test.
FR-104	Intake	Suitability	Check temperature, container, and elapsed time against rules.	Should	**Rationale:** Prevent testing degraded samples. **Verify:** Test edge metrics.
FR-105	Tracking	Splits	Support sample splits and link to parent reference key.	Must	**Rationale:** Mandatory for external referrals. **Verify:** Inspect DB relations.
FR-106	Tracking	Immutable Log	Add log row automatically when location or staff ID changes.	Must	**Rationale:** Eliminate mix-ups and log failures. **Verify:** Delete/update blocking.
FR-107	Lab Ops	Segregation	Separate raw data from reportable fields until review is done.	Must	**Rationale:** Prevent customer viewing unverified values. **Verify:** Portal isolation test.
FR-108	Lab Ops	Validation	Verify analyst ID against certified methods list before save.	Should	**Rationale:** Quality assurance check. **Verify:** Uncertified save block.
FR-109	Lab Ops	Machine Block	Set test to 'Delayed' if machine fails QC check or is out of service.	Must	**Rationale:** Prevent execution on uncalibrated hardware. **Verify:** Status shift demo.
FR-110	Reports	Version Ctrl	Increment version number and force revision notes during corrections.	Must	**Rationale:** Vital for audit trails and liability protection. **Verify:** Blank note block test.
FR-111	Reports	Dashboard	Show unapproved corrected reports, anomalies, referrals, and delays.	Should	**Rationale:** Live operational visibility for managers. **Verify:** UI rendering check.
NFR-201	Offline	Sync Cap	Store 200 logs offline and sync in under 30 seconds.	Must	**Rationale:** Counteract weak rural networks. **Verify:** Cache sync simulation.
NFR-202	Security	Multi-Tenant	Filter queries by sensitivity flag; return HTTP 403 error.	Must	**Rationale:** Absolute multi-tenant isolation. **Verify:** Token injection attempt.
NFR-203	Alerts	Latency	Send email/SMS within 5 minutes when Rejected or Delayed.	Should	**Rationale:** Improve response speed. **Verify:** Timestamp log check.
NFR-204	Scope	Isolation	Zero software dependencies on billing or enforcement modules.	Must	**Rationale:** Enforce defined brief boundaries. **Verify:** Architecture tree inspection.

5.2 Business Rules and Constraints

• **Role-Based Testing:** Staff can assign tests to analysts only if the analyst has recorded training for that method.

• **Mandatory Data Separation:** The system must separate general text descriptions from precise GPS data in the database.

• **Release Hierarchy:** Raw machine data are provisional values. The system must block the release of provisional values to customers until a Quality Manager approves the report.

• **Data Access Restrictions:** When a customer record has an active commercial sensitivity flag, the system must hide these test results from all other customers.

5.3 Assumptions and Constraints

• **Provisional Sample Lifespan:** The system assumes a maximum holding time of 48 hours for provisional samples before automatic rejection.

• **Component-Level Suitability:** If one test fails sample suitability checks, other compliant tests on the same sample can stay active.

• **Chain of Custody Inheritance:** Sample portions automatically inherit the primary Laboratory Reference ID as a parent reference.

• **Unreliable Connectivity:** Internet access is stable at the Tralee office but weak at rural collection points. The field module must support offline data entry.

5.4 System Risks

• **Traceability Breakdowns:** Complex custody processes can cause staff to bypass the digital system. This leads to continued sample mix-ups.

• **Premature Machine Integration:** Direct connections to laboratory hardware APIs can cause project schedule delays.

• **Data Siloing Failures:** Incorrect data access controls can allow unauthorized users to see sensitive customer data.

5.5 Clarification Questions

• What is the maximum time a sample can stay provisionally accepted before automatic rejection?

• Is there a matrix that maps specific container defects directly to the tests they invalidate?

• Which specific roles can authorize repeat tests, method departures, or corrected reports?

• Should updated completion dates notify the customer immediately or wait for manager review?

• What specific fields should be visible to partner organizations versus field technicians?

6. Behavioural & Use-Case Modelling

6.1 Use-Case Specifications

UC-101: Register Sample & Verify Suitability

Actor: Reception Technician

Preconditions: The system is online. A physical sample arrives at the Tralee laboratory. A sample request exists in the system.

Trigger: The technician starts the intake workflow on the system.

Basic Flow:

1. The technician selects the correct request ID.
2. The technician selects the sampling location and the precise GPS Sampling Point.
3. The technician inputs sample type, collection time, collector ID, container type, receipt temperature, and volume.
4. The system generates a unique laboratory reference ID.
5. The system evaluates the sample metrics against the suitability matrix rules.
6. The metrics pass the rules. The system sets the status to "Accepted".
7. The system creates the first row in the Chain of Custody log. It sets the location to "Reception".
8. The system shows a confirmation message to the user.

Alternative Flows:

AF-1: Data is Incomplete or Near Thresholds:

1. The system prompts the user for receipt condition notes.
2. The technician inputs the specific defects of the data.
3. The system sets the status to "Provisionally Accepted".
4. The system starts a 48-hour data rejection countdown timer.
5. The flow continues at Step 7 of the basic flow.

AF-2: Split Sample (Parent-Child relation):

1. The operator selects the split command.
2. The system creates a sub-sample record.
3. The system copies the original ID to the parent reference field.
4. If the test is external, the system sets the external referral flag to true and records the external laboratory ID.

Exception Flows:

EF-1: Critical Defect Found (Leak, Expired Time, or Wrong Container):

1. The system detects a critical rule failure.
2. The system sets the status to "Rejected" and locks the sample record.
3. The notification subsystem sends an alert to the contact list within 5 minutes.
4. The process stops.

Postconditions:

Success: The system saves the sample with a unique ID. The status is "Accepted" or "Provisionally Accepted". The chain of custody log starts.

Failure: The status is "Rejected". The system blocks test assignments and notifies the customer.

6.2 Gherkin Feature Specifications

```
Feature 1: Unique Identification and Coordinate Mapping
  Rule: Every logged sample must have a unique identifier and spatial mapping
    Scenario: Successful sample intake under standard conditions
      Given a physical sample has arrived at the laboratory intake desk
      And a valid sample request exists in the system
      When the Reception Technician enters valid intake metrics and selects a precise GPS Sampling Point
      Then the system should generate a unique laboratory reference ID
      And set the sample status to "Accepted"
      And append an initial row to the Chain of Custody log with the location set to "Reception"

Feature 2: Offline Field Collection
  Rule: Field applications must support offline operations and rapid data synchronization
    Scenario: Capture sampling coordinates during cellular network disconnection
      Given the mobile field application detects that network connectivity is unavailable
      When the Field Technician saves the GPS coordinates and collection data for a new sample
      Then the system should store the collection log line securely within the local device cache

Feature 3: Sample Splitting and Custody Tracking
  Rule: Sample portions must inherit parent identity, and custody logs must be immutable
    Scenario: Creating a child sample fraction for external referral testing
      Given an environmental sample exists in the system with an active status
      When an Analyst executes a split command to create a sub-sample fraction for an external test
      Then the system should automatically link the parent sample reference ID to the new fraction
      And set the external referral flag to true
```

7. Domain Analysis & Conceptual Model

7.1 Candidate-Concept Filtering Rationale

Extracted Noun	Category	System Decision	Rationale
Customer	Person/Role	KEEP	Core structural user requiring high data isolation profiles.
Field Technician	Person/Role	KEEP	Primary operational actor generating remote coordinates.
Handwritten Label	Attribute	REMOVE	Replaced entirely by the system's digital ID logic.
Environmental Sample	Physical Entity	KEEP	Central business concept; holds state, volume, and material type.
Sample Portion	Physical Entity	KEEP	Essential split object linking directly to a primary parent ID.
Bench Sheet	Logical Entity	MERGE	Merged inside the structural layout of the Custody Log entity.
Hardware API Driver	Design Element	REMOVE	Explicitly out-of-scope based on baseline boundary conditions.
Provisional Value	Data Attribute	KEEP	Must be held in separate data segments away from final views.

7.2 Domain Glossary

Customer: Person or group that requests testing. Relates to Sample Request and Approved Report.

Field Technician: Operator who collects field samples. Relates to Sampling Point and Environmental Sample.

Reception Technician: Worker who receives samples at lab. Relates to Laboratory Reference and Suitability.

Analyst: Specialist who performs lab testing. Relates to Requested Test, Calibration, and QC Check.

Quality Manager: Person who verifies data and reports. Relates to Reportable Result and Approved Report.

Laboratory Manager: Supervisor who monitors lab operations. Relates to External Lab and Operations Dashboard.

Sample Request: Submission documenting requested tests. Relates to Customer and Environmental Sample.

Sampling Location: Broad description of where sample is taken. Relates to Sampling Point and Sample Request.

Sampling Point: Exact GPS coordinates of collection. Relates to Sampling Location and Field Technician.

Environmental Sample: Water or soil quantity meant for testing. Relates to Sampling Point and Sample Portion.

Sample Portion: Split or fraction of a parent sample. Relates to Environmental Sample and External Lab.

Laboratory Reference: Unique ID assigned at intake. Relates to Reception Technician and Sample.

Chain of Custody: Chronological record of sample transfers. Relates to Bench Sheet and Laboratory Reference.

Requested Test: Analytical test chosen by customer. Relates to Sample Request and Analyst.

Instrument Calibration: Adjustment process using standard controls. Relates to Analyst and Quality-Control Check.

Quality-Control Check: Metric run to verify machine accuracy. Relates to Calibration and Reportable Result.

Provisional Value: Calculated value awaiting technical review. Relates to Observation and Reportable Result.

Reportable Result: Final data reviewed and approved for release. Relates to Provisional Value and Quality Manager.

Approved Report: Final authorized compilation for customer. Relates to Reportable Result and Quality Manager.

Corrected Report: Replacement report tracking revisions. Relates to Approved Report and Quality Manager.

8. Traceability & Project Execution

8.1 Recommended Repository Structure

```
ketco-udp-analysis/
■■■ README.md
■■■ report/
■ ■■■ final-specification-report.md
■■■ requirements/
■ ■■■ functional-requirements.md
■ ■■■ non-functional-requirements.md
■■■ features/
■ ■■■ 01-intake-and-mapping.feature
■ ■■■ 02-offline-field-collection.feature
■ ■■■ 03-chain-of-custody.feature
■ ■■■ 04-lab-operations-and-instruments.feature
■ ■■■ 05-data-privacy-and-segregation.feature
■ ■■■ 06-versioned-reports.feature
■■■ diagrams/
■ ■■■ use-case-model.puml
■ ■■■ domain-model.puml
■■■ decisions/
■ ■■■ ADR-001-offline-data-caching.md
■ ■■■ ADR-002-data-segregation-architecture.md
■■■ references/
    ■■■ asd-ste100-guidelines.pdf
    ■■■ iso-17025-compliance-notes.txt
```

9. Reflection & AI Declaration

9.1 Critical Design Decisions & Revisions

• **Permanent Custody Records:** Users cannot change or delete data in the Chain of Custody log. This lock stops data tampering and keeps a full audit trail.

• **Separation of Location Data:** The system separates broad text locations from precise GPS data points. This rule prevents geographic errors during data mapping.

• **Controlled Data Visibility:** The system stores provisional values in separate data blocks from final results. This protection stops customers from seeing unverified data before the quality review.

9.2 Limitations & Unresolved Questions

• **Material Data Validation:** Laboratory managers must verify the numeric values that connect container materials to soil test limits.

• **Access Permissions:** The project team must define the data fields that partner organizations can see during transit if rules change.

9.3 AI Level Disclosure & Interaction Log

AI Declaration: Generative AI tools supported document formatting, structure verification, and ASD-STE100 layout adaptation. All core domain engineering parameters, architectural scoping, and boundary assertions were drafted independently by human analysis.

Checked and Rejected AI Suggestion Example:

AI Suggestion: During an internal optimization critique, the AI system recommended adding an automated alert mechanism to notify state environmental authorities whenever an Unusual Result flag was triggered.

Human Modification/Correction: This suggestion was explicitly rejected. The known business constraints state that an unusual result is a statistical anomaly and does not prove environmental pollution. Furthermore, Section 3.2 explicitly declares automated regulatory enforcement alerts completely out-of-scope. Adding this feature would have introduced severe scope creep and legal liability risks for KETCo.